Recursive vs. Iterative Palindrome Check

The first algorithm will use an **iterative approach**, the second algorithm will use a **recursive approach**.

Iterative Approach

An iterative approach is based around the use of a **loop** which can be:

- A count-controlled loop (e.g. *FOR loop*)
- A condition-controlled loop (e.g. *WHILE loop* or *REPEAT UNTIL loop*)

For our iterative palindrome check algorithm, we will use a loop to check all the letters in the first half of the word and compare them with the letters in the second half of the word (in reverse order). If they all match then the word is a palindrome.

Recursive Apporach

A recursive function is a function that:

- Includes a **call to itself**,
- Has a **stopping condition** to stop the recursion.

For our recursive palindrome check algorithm, we will use a function that:

- Checks that the word is at least two characters long:
- If the word is less than two characters then the function will stop the recursion (**stopping condition**) and Return True as the word is a palindrome.
- If the word is two or more characters long, the function will check that the first and last letter of the word are the same:
- If they are the same, the function will extract the word in the middle (remove the first and the last letter) and **call itself** with this new shortest word.
- If they are different, then the word is not a palindrome. The function will stop the recursion here (**stopping condition**) and return False.

Algorithm

1. Start
2. Declare a string variable.
3. Ask the user to initialize the string.
4. Call a function to check whether the string is palindrome or not.
5. If a string is empty, then it is a palindrome.
6. If the string is not empty, then call a recursive function.
7. If there is only one character, then it is a palindrome.

8. If the first and last characters do not match, then it is not a palindrome.
9. If there are multiple characters, check if the middle substring is also palindrome or not using the recursive function.
10. Print the result.
11. Stop.

```
Enter the String: wow
wow is palindrome
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```
Enter the String: hello
hello is not a palindrome
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